

Manufacturing Batch-Review

A Governed Agentic AI Reference Architecture for Life Sciences

Agent 09 of 8 · review electronic batch records by exception — QA owns every release/reject

HCLS AI Agent Suite

HCLS AI Agent Suite · Built on AWS · June 2026

FROM 48-HOUR REVIEWS TO REVIEW BY EXCEPTION

Manufacturing Batch-Review

A governed batch-review copilot that reads the electronic batch record and process data, flags deviations and out-of-spec results by exception, and drafts the disposition — while a QA reviewer makes and signs every release/reject decision.

62%

of US drug shortages trace to manufacturing/quality problems — the #1 root cause

[gov/peer-reviewed — FDA]

~48 hrs

average review per batch report today (up to ~500 hrs for complex paper-based)

[sector-press/estimate]

~\$14K

average cost per deviation investigation (mostly senior labor)

[sector-press/estimate]

The Issue & The Cost of Doing Nothing

THE ISSUE

- Batch-record review and OOS/deviation investigation are slow, manual, and senior-labor-heavy.
- 21 CFR 211.192 (production-record review / discrepancy investigation) is among the most-cited FDA warning-letter findings.
- Right-first-time for complex biologics often sits ~80%; every non-RFT batch triggers rework and release delay.
- A single biologics batch failure can cost tens of millions; raw materials alone often exceed \$1-2M.

THE COST OF DOING NOTHING

~\$420K / yr

Modeled: 200 batches/yr × ~15% deviating (85% RFT) ≈ 30 investigations × ~\$14K labor ≈ ~\$420K/yr — before any scrapped batch (\$1-2M+ each) or delayed-release carrying cost.

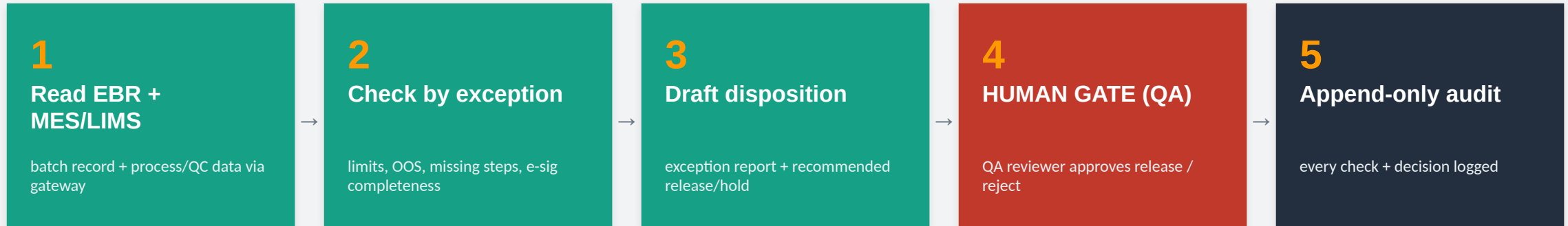
- Release delays and drug-shortage risk from review backlogs.
- Data-integrity / 211.192 findings from inconsistent record review.

[modeled — deviation-investigation cost × batch volume]

The design line: the agent reviews records and flags exceptions; a QA reviewer makes and signs the final release/reject decision — the agent never releases a batch.

How We Solve It — A Governed Pipeline

Read the electronic batch record + MES/LIMS data through a scoped gateway, check by exception and draft the disposition — then stop at a QA gate before release.



Every action audited

node, timestamp, actor, data sources, model + prompt version — append-only and inspection-ready (21 CFR Part 11).

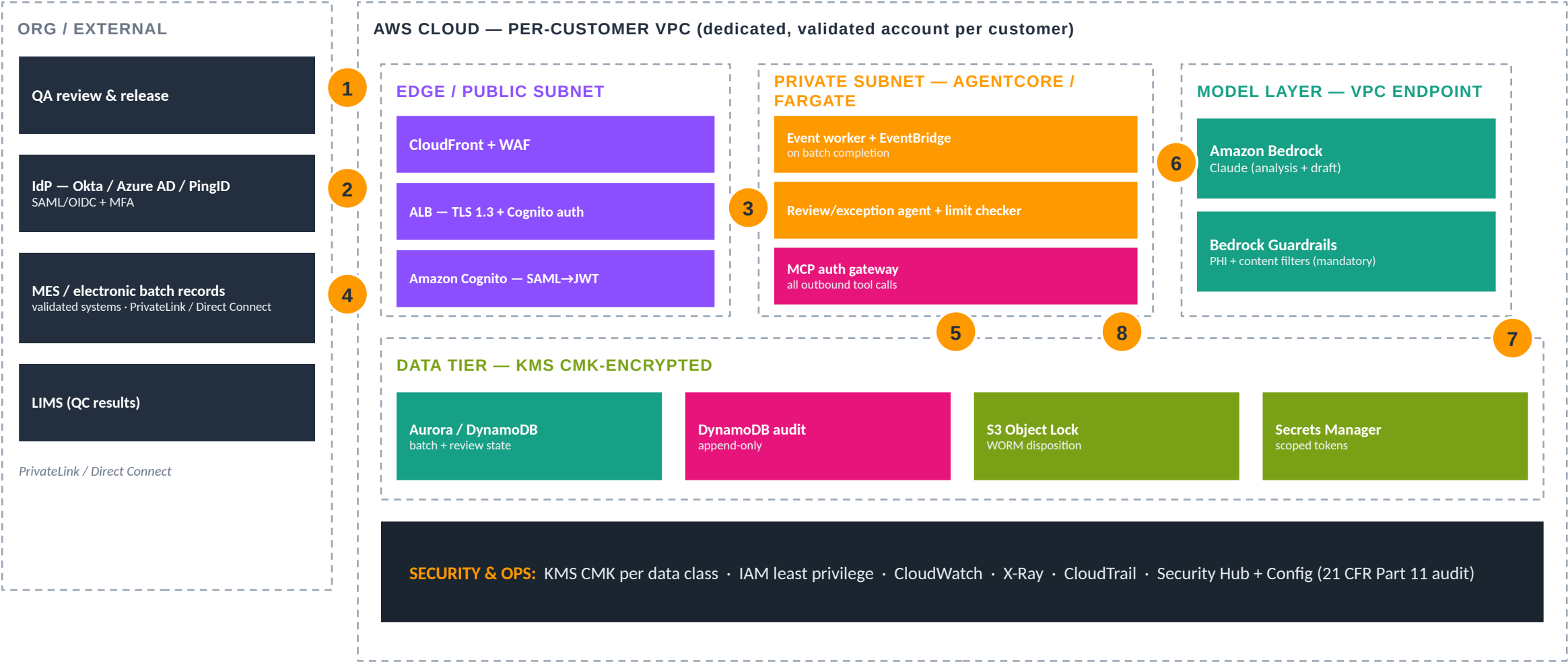
Grounded & explainable

every regulated claim traces to the source corpus; Bedrock Guardrails mask PHI before any model call and bound the output.

AI never decides batch release

the consequential action is a human-only step, enforced in the orchestration framework — not a policy SOP.

AWS Architecture & Traffic Flow



1 QA sign-in 2 SAML federation (MFA at IdP; Cognito issues JWT) 3 authenticated session to review console 4 MES/EBR + LIMS reads over private connectivity 5 EventBridge triggers review on batch completion 6 model calls stay in-VPC via Bedrock + Guardrails 7 every flag, exception, and release decision persisted to append-only audit (21 CFR Part 11) 8 connectors reachable only through the governed MCP gateway

Proof, Payback & How to Deploy

MEASURED OUTCOMES

~80%

batch-release time cut with review by exception

[sector-press/estimate]

#2 cited

211.192 record-review is a top FDA warning-letter finding

[gov/peer-reviewed]

>50%

deviation reduction at a benchmarked biopharma site

[industry-research — McKinsey]

36 tests

passing with no API key (built to flagship depth)

[built]

WHAT IT TAKES TO DEPLOY

1. Provision KMS CMK + per-customer validated VPC / network.
1. Stand up Cognito SAML→JWT federation (operator / QA-reviewer roles).
1. Front with CloudFront + WAF, ALB TLS 1.3.
1. Deploy the agent + grant tools via the MCP gateway (deny-by-default).
1. Connect MES / electronic batch records + LIMS; wire EventBridge on batch completion.
1. Enable S3 Object Lock + append-only audit; run smoke + QA-gate test.

Deploy: CloudFormation quick-deploy provisions the isolated environment + dual MCP gateway + connector Lambdas (mes/lims); connect MES/EBR + LIMS. AWS-native Strands + Step Functions rebuild included.

aws-native-reference/09-manufacturing-batch-review/DEPLOY.md · docs/DEPLOY-QUICKSTART.md

The takeaway: the agent isn't the product — the governed platform that makes it 21 CFR Part 11 / GxP-defensible and deployable on AWS is.